

Cross Section

$$\frac{dN}{dt} = \sigma \cdot \frac{dN_p}{dt} \cdot n_b d$$

colliders:

$$\frac{dN}{dt} = \sigma \cdot \mathcal{L}$$

$\mathcal{L}$: inst. luminosity of collider.

$$N_{\text{events}} = \int \sigma \cdot \mathcal{L} \cdot dt = \sigma \cdot \int \mathcal{L} \cdot dt$$

$\mathcal{L}$: integrated luminos.

$$[\sigma] = \text{cm}^2$$

1 barn. $\Rightarrow$ $\mathcal{L}$ measured in b^{-1}

$$N_{\text{phys}} = \sigma \cdot \mathcal{L}$$

$\mathcal{L} \rightarrow$ parameter of the collider
 $\sigma \rightarrow$ physics of the process

σ : depends on

- nature of particles interacting
- their energy.
- their physical param.

$$\sqrt{s} = \sqrt{(P_1 + P_2)^2}$$

LHC: sym. collider of protons.

$$P_1 = (E, \vec{p})$$

$$P_2 = (E, -\vec{p})$$



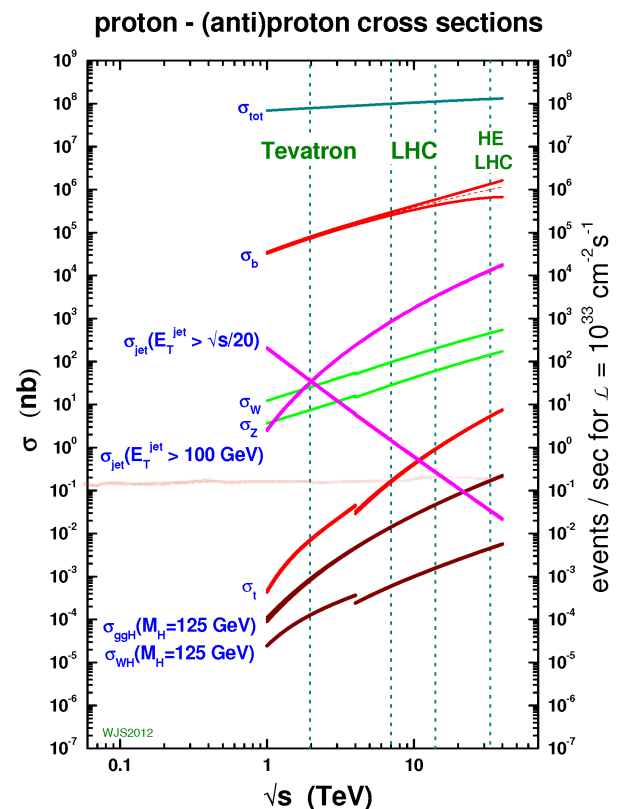
$$P_1 + P_2 = (2E, \vec{0})$$

$$\sqrt{s} = 2E$$

E : energy of the beam

Total cross section

$p+p \rightarrow \text{anything}$



$$\begin{aligned}
 p+p &\rightarrow p+p \\
 p+p &\rightarrow p+p+\pi^+\pi^- \\
 p+p &\rightarrow p+p+p+\bar{p} \\
 &\quad p+p+\pi^0 \\
 &\quad \vdots \\
 &\quad H+X \\
 &\quad X_{\text{new}}+\bar{X}_{\text{new}}
 \end{aligned}$$

$$E = 6.8 \text{ TeV}$$

$$\sigma_{\text{tot}} = 10^8 \text{ nb} \approx 10^{-1} \text{ b}$$

$$\sigma_b: p+p \rightarrow b\text{quark} + X$$

$$\sigma_t: p+p \rightarrow \text{top} + X$$

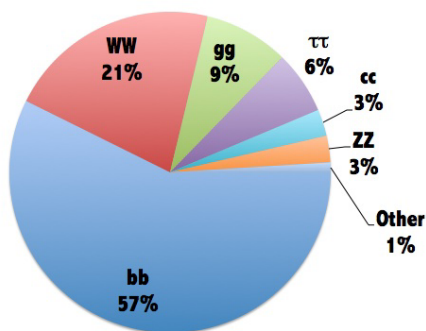
$$\sigma_H: p+p \rightarrow \text{Higgs} + X$$

inclusive cross section for Higgs production

$$\sigma_H(\sqrt{s} = 13. \text{ TeV}) \approx 10^{-1} \text{ nb}$$

$$H \rightarrow Z^+ Z^-$$

Higgs decays at $m_H=125\text{GeV}$



Z DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
$\Gamma_1 e^+e^-$	(3.363 $\pm$ 0.004) %	
$\Gamma_2 \mu^+\mu^-$	(3.366 $\pm$ 0.007) %	
$\Gamma_3 \tau^+\tau^-$	(3.370 $\pm$ 0.008) %	
$\Gamma_4 \ell^+\ell^-$	[a] (3.3658 $\pm$ 0.0023) %	
Γ_5 invisible	(20.00 $\pm$ 0.06) %	
Γ_6 hadrons	(69.91 $\pm$ 0.06) %	
$\Gamma_7 (u\bar{u}+c\bar{c})/2$	(11.6 $\pm$ 0.6) %	
$\Gamma_8 (d\bar{d}+s\bar{s}+b\bar{b})/3$	(15.6 $\pm$ 0.4) %	
$\Gamma_9 c\bar{c}$	(12.03 $\pm$ 0.21) %	
$\Gamma_{10} b\bar{b}$	(15.12 $\pm$ 0.05) %	
$\Gamma_{11} b\bar{b}b\bar{b}$	(3.6 $\pm$ 1.3) $\times 10^{-4}$	
$\Gamma_{12} gg$	< 1.1 %	CL=95%
$\Gamma_{13} \pi^0\gamma$	< 5.2 $\times 10^{-5}$	CL=95%
$\Gamma_{14} \eta\gamma$	< 5.1 $\times 10^{-5}$	CL=95%
$\Gamma_{15} \omega\gamma$	< 6.5 $\times 10^{-4}$	CL=95%
$\Gamma_{16} \eta'(958)\gamma$	< 4.2 $\times 10^{-5}$	CL=95%
$\Gamma_{17} \gamma\gamma$	< 5.2 $\times 10^{-5}$	CL=95%
$\Gamma_{18} \gamma\gamma\gamma$	< 1.0 $\times 10^{-5}$	CL=95%
$\Gamma_{19} \pi^\pm W^\mp$	[b] < 7 $\times 10^{-5}$	CL=95%

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$$\begin{aligned}
 &\text{cross section} \\
 p+p &\rightarrow H+X \\
 &\quad \rightarrow Z^0+Z^0 \\
 &\quad \quad \rightarrow \mu^+\mu^- \\
 &\quad \quad \rightarrow e^+e^-
 \end{aligned}$$

decay width.
Branching fraction

$$N(p\bar{p} \rightarrow H + X \rightarrow Z^0 Z^0 \rightarrow \mu^+ \mu^- e^+ e^-) =$$

$$\sigma_H \cdot L \cdot \text{BF}(H \rightarrow Z Z) \times \text{BF}(Z \rightarrow e e) \times \text{BF}(Z \rightarrow \mu \mu) \times 2$$

Branching fraction:

unstable particle Γ : total decay width $\Gamma_{\text{tot}} \approx \frac{1}{\tau}$

$$H \rightarrow \begin{array}{c} Z Z \\ \gamma \gamma \\ b \bar{b} \\ W W \\ c \bar{c} \\ Z \nu \end{array} \quad \begin{array}{c} \Gamma(H \rightarrow Z Z) \\ \Gamma(H \rightarrow \gamma \gamma) \\ \cdot \\ \cdot \\ \cdot \end{array}$$

$$\Gamma_{\text{tot}} = \sum_i \Gamma(H \rightarrow \text{final state } i)$$

$$\text{BF}(H \rightarrow \text{state } i) := \frac{\Gamma(H \rightarrow \text{state } i)}{\Gamma_{\text{tot}}}$$

$$N(p\bar{p} \rightarrow H \rightarrow e e \mu \mu + X) = \sigma_H \cdot L \cdot 3 \times 10^{-2} \cdot 3 \times 10^{-2} \cdot 3 \times 10^{-2} \times 2$$

$$\approx \sigma_H \cdot L \cdot 50 \times 10^{-6}$$

$$\sigma_H \approx 15 \text{ fb}$$

$$= 10^8 \text{ nb}$$

$$\approx L \cdot 10^8 \text{ nb} \cdot 50 \times 10^{-6}$$

$$\approx \textcircled{L} \cdot 10^3 \cdot 5 \cdot (\text{nb})$$

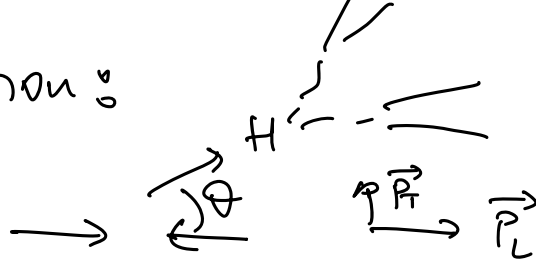
$$L = 1 \text{ nb}^{-1}$$

$$N = 10^3 \cdot 5 = 5000 \text{ candidates produced.}$$

$$\text{in 2014: } L \approx 120 \text{ fb}^{-1} = 120 \times 10^6 \text{ nb}^{-1}$$

σ_{tot} : total number of events.

Differential cross section:

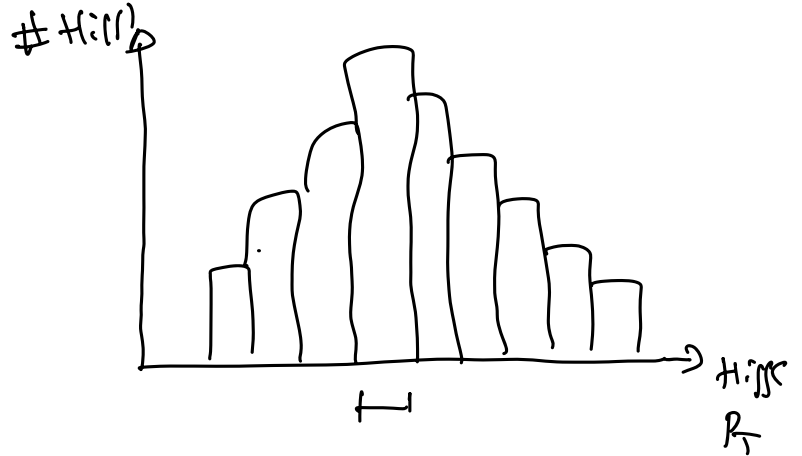


Higgs produced at different angles.
different momentum

$$\frac{d\sigma}{dP_T} \propto \frac{\# \text{ events}}{dP_T}$$

Differential cross section.

$$\frac{dN}{dP d\Omega}$$

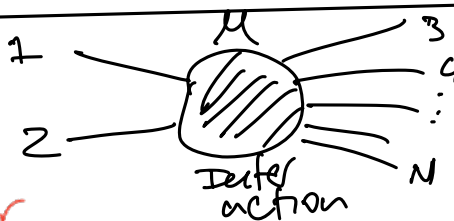


$$d\Omega = d\cos\theta \cdot d\phi$$

$$\frac{dN}{d\theta}, \frac{dN}{d\cos\theta}$$

$\frac{d\sigma}{d\Omega}$ is one of the tools to study collisions

Fermi Golden Rule for Scattering



$$1+2 \rightarrow 3+4+\dots+N$$

Conservation of Energy-momentum

$$\sigma = \frac{S}{4\sqrt{(\underline{P}_1 \cdot \underline{P}_2)^2 - (m_1 m_2)^2}} \int |\mathcal{M}|^2 (2\pi)^4 \delta^4(\underline{P}_1 + \underline{P}_2 - \underline{P}_3 - \underline{P}_4 - \dots - \underline{P}_N) \times$$

relative velocity between 1 and 2

$$\times \prod_{j=3}^N \frac{1}{2\sqrt{\underline{P}_j^2 + m_j^2}} \frac{d^3 \underline{P}_j}{(2\pi)^3}$$

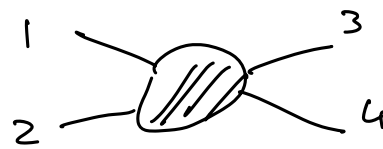
$$\underline{P}_j = |\underline{P}_j|$$

phase space in final state

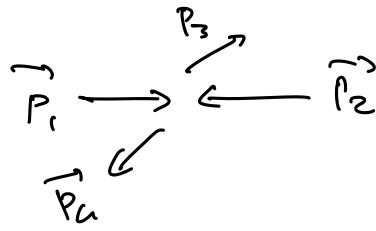
$$S: a+b \rightarrow c+d+\dots$$

$$S = \frac{1}{n_c!} \frac{1}{n_d!}$$

Example: $1+2 \rightarrow 3+4$



center of mass:



$$\vec{P}_1 + \vec{P}_2 = \vec{0} = \vec{P}_3 + \vec{P}_4$$

$$\underline{P}_1 = (E_1, \vec{P}_1)$$

$$\underline{P}_2 = (E_2, -\vec{P}_1)$$

$$\sigma = \frac{1}{4 \sqrt{|\vec{P}_1|^2 (E_1 + E_2)^2}} \int |\mathcal{M}|^2 (2\pi)^4 \delta(E_1 + E_2 - E_3 - E_4) \delta^3(\vec{P}_1 + \vec{P}_2 - \vec{P}_3 - \vec{P}_4)$$

$$\vec{P}_3 = -\vec{P}_4 = \vec{P}_{out}$$

$$\vec{P}_1 = -\vec{P}_2 = \vec{P}_{in}$$

$$\frac{1}{2\sqrt{P_3^2 + m_3^2}} \frac{1}{2\sqrt{P_4^2 + m_4^2}} \frac{d^3\vec{P}_3}{(2\pi)^3} \frac{d^3\vec{P}_4}{(2\pi)^3}$$

$$E_1 = \sqrt{P_{in}^2 + m_1^2}$$

$$E_3 = \sqrt{P_{out}^2 + m_3^2}$$

$$d^3\vec{P}_3 \equiv d^3\vec{P}_{out} = P_{out}^2 dP_{out} d\Omega$$

$$\frac{d\sigma}{d\Omega} = \frac{S}{64\pi^2} \frac{1}{(E_1 + E_2)^2} \frac{|\vec{P}_{out}|}{|\vec{P}_{in}|} |\mathcal{M}|^2$$